

Labdhi Shah

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Education

Thakur College of Engineering and Technology, Mumbai
B.Tech in Artificial Intelligence and Data Science

July 2023 – Expected July 2027
CGPA: 8.5

Work Experience

Miscot System Pvt. Ltd.

December 2025 – February 2026

Software Developer Intern

Mumbai

- Built an **OCR pipeline** for printed documents using **PaddleOCR** with DBNet text detection and PARSeq recognition; benchmarked it against DocTR on accuracy and performance, and tuned image preprocessing (denoising, binarization, normalization) to improve input quality.
- Added **YOLO-based field detection**, **Aadhaar masking** and signature-region cropping to the document pipeline.
- Built **FastAPI** endpoints exposing the ML and document-processing pipelines to other services.
- Built a **hybrid retrieval chatbot** that moved from TF-IDF/fuzzy matching to **embedding-based retrieval**, with a similarity threshold routing between rule-based answers and an LLM API.

Projects

Knowledge Assistant — Document Q&A with Verified Citations | Python, FastAPI, Next.js, PostgreSQL

[GitHub](#)

- Built and deployed a **full-stack RAG assistant** that answers questions about uploaded PDFs, Word documents and source files, citing the exact page and section behind every claim; runs on Vercel as two services with a private backend.
- Built **layout-aware PDF extraction** from font sizes and text geometry, and a sentence-boundary chunker that carries section headings forward and records exact source offsets for traceable citations.
- Evaluated retrieval on a 28-question labelled test set — **83% top-3 accuracy, 96% top-10, under 500 ms per query**; A/B-tested an alternative chunking strategy on the same ground truth and rejected it on the evidence (78% vs 83%).
- Prevented fabricated answers: thresholds derived from measured score distributions decline low-evidence questions **before any model call**, and citations to passages that were not retrieved are stripped after generation.
- Extended to summarization, document comparison, multi-turn follow-ups and per-request choice between Claude and Gemini; measured **15/15 task-routing accuracy** and **zero citation errors** on both providers.

L-Mart — AI Loyalty Operations Agent | Python, FastAPI, PostgreSQL/Supabase, pgvector, Next.js, MCP, Vercel

[GitHub](#)

- Built an agent that answers open-ended retail-loyalty questions by planning and running tool calls; implemented **tool calling without an agent framework**, with one registry enforcing schemas, permissions and dispatch for 10 tools. The model parameterises 11 versioned SQL metrics but never writes SQL.
- Modelled the agent loop as a **durable Postgres state machine**, one model turn per HTTP request, so runs survive serverless timeouts and resume across requests; a lease-based turn claim stops retries from running a turn twice.
- Built **hybrid retrieval** over a 22-document policy corpus, fusing Postgres full-text search and pgvector with Reciprocal Rank Fusion; raised **top-3 recall from 73% to 96%** by fixing jurisdiction filtering and capping chunks per document.
- Exposed the same tool registry over **Model Context Protocol** (stdio, Streamable HTTP) with one authorization path shared by the agent and external clients.
- Required **named human approval**, bound to a content hash, before any campaign reaches a customer, backed by role-based access, idempotent execution and an append-only audit log; **200+ automated tests**, none calling a model.

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, HTML/CSS

Backend & Databases: FastAPI, Flask, Pydantic, REST APIs, PostgreSQL, Supabase, MySQL, Next.js, React

AI/ML & LLMs: LLM tool calling, AI agents, MCP, prompt engineering, prompt caching, LLM evaluation, Claude / Gemini / OpenAI-compatible APIs, Scikit-learn, Pandas, NumPy

Retrieval & RAG: Embeddings, vector search (pgvector, FAISS), hybrid search (full-text + vector, RRF), TF-IDF, fuzzy matching, document chunking

Document AI & Vision: OCR (PaddleOCR, DocTR, DBNet, PARSeq), YOLO, OpenCV, image preprocessing

Tools & Deployment: Git, GitHub, pytest, Vercel, serverless deployment

Achievements & Leadership

2nd Place — IndiaNext Hackathon (National Level)

March 2026

- Placed **2nd of 234 teams** in a 24-hour hackathon (team of 4) with CarbonGrid, an AI sustainability platform for net-zero planning.

Data Science Club (S4DS)

August 2024 – September 2025

- Led a **100+ member** student community; organized technical workshops and events with **300+ attendees**.